

Cyber Security Task 02: CIA Triad & Risk Assessment

Part A: Understanding the CIA Triad

Research and explain:

1. Confidentiality

Definition: Confidentiality means keeping information private and making sure only authorized people can access it.

Why it is important: It protects personal and sensitive information from being seen or used by the wrong people.

Real-world example: It protects personal and sensitive information from being seen or used by the wrong people.

2. Integrity

Definition: Integrity means keeping information accurate, complete, and unchanged unless authorized changes are made.

Why it is important: It helps ensure that information can be trusted and is not altered by mistakes or unauthorized people

Real-world example: A student's exam marks stored in a school database should not be changed by anyone except authorized staff. This keeps the records accurate and reliable.

3. Availability

Definition: Availability means ensuring that information, systems, and services are accessible whenever authorized users need them.

Why it is important: It allows people to access important data and services without unnecessary delays or interruptions.

Real-world example: An online banking app should be available 24/7 so customers can check their account balance or transfer money whenever they need to.

Comparison Table

Aspect	Confidentiality	Integrity	Availability
Definition	Keeping information private and accessible only to authorized people.	Ensuring information remains accurate and unchanged unless authorized.	Making sure information and systems are accessible when needed.
Why it is Important	Protects sensitive information from unauthorized access.	Ensures information is trustworthy and reliable.	Allows users to access data and services without interruption.
Real-world Example	Using a password to protect an email account.	School exam records can only be changed by authorized staff.	An online banking app is available for customers at any time.

Part B: Real-World Case Study

Choose any one cyber attack from the following:

- WannaCry Ransomware
- Equifax Data Breach
- SolarWinds Attack
- Colonial Pipeline Attack
- Facebook Data Leak

WannaCry Ransomware

- **What happened?**
 - WannaCry was a ransomware attack that spread across the world in 2017. It infected computers, locked users' files, and demanded money to unlock them.
- **Which part of the CIA Triad was affected?**
 - WannaCry was a ransomware attack that spread across the world in 2017. It infected computers, locked users' files, and demanded money to unlock them.
- **What was the impact?**
 - The attack affected hospitals, businesses, and government organizations in many countries. Important services were disrupted, and many organizations lost time and money.
- **How could it have been prevented?**
 - The attack could have been prevented by regularly updating software, installing security patches, using antivirus programs, and backing up important data.

Part C: Risk Assessment Activity

Risk Assessment table

Asset	Threat	Impact	Risk Level
Student Records	Data Breach	High	Critical
Examination Data	Unauthorized Access or Tampering	High	High
College Website	Website Hacking or DDoS Attack	High	High
Faculty Information	Identity Theft or Data Leak	Medium	Medium
Network Infrastructure	Malware or Ransomware Attack	High	Critical

Part D: Threat Identification

Research and explain the following threats:

- **Insider Threat**

Description: An insider threat occurs when a person within an organization, such as an employee, student, or staff member, intentionally or accidentally misuses their access to systems, data, or resources. Since insiders already have authorized access, they can cause significant damage if security measures are not followed.

Possible Impact:

- Data theft or leakage
- Loss of confidential information
- Financial and reputational damage

Prevention Methods:

- Limit access to sensitive data
- Monitor user activities
- Provide cybersecurity awareness training

- **Malware**

Description: Malware is any software that is intentionally designed to harm a computer, network, or user. It includes viruses, worms, spyware, ransomware, and trojans. Malware can enter a system through infected downloads, malicious websites, email attachments, or removable storage devices.

Possible Impact:

- Loss of important data
- System damage or slowdown
- Theft of personal and financial information

Prevention Method:

- Install and update antivirus software
- Avoid downloading files from unknown sources
- Keep software and operating systems updated

- **Phishing**

Description: Phishing is a cyberattack in which attackers pretend to be a trusted person or organization to trick users into revealing sensitive information such as passwords, bank details, or personal data. Phishing attacks commonly occur through emails, text messages, or fake websites that look legitimate.

Possible Outcome:

- Stolen passwords and account credentials
- Financial loss
- Identity theft and unauthorized access

Prevention Methods:

- Verify suspicious emails and links before clicking
- Avoid sharing personal information online
- Enable two-factor authentication

- **Weak Passwords**

Description: Weak passwords are passwords that are short, simple, or easy to guess. Examples include common words, names, birthdays, or number sequences. Attackers can easily crack weak passwords using automated tools, giving them unauthorized access to accounts and systems.

Possible Outcome:

- Unauthorized account access
- Data theft or modification
- Compromised personal and organizational information

Prevention Methods:

- Use strong and unique passwords
- Combine letters, numbers, and special characters
- Change passwords regularly and avoid reusing them

- **Unpatched Systems**

Description: Unpatched systems are computers, applications, or devices that have not been updated with the latest security patches released by software developers. These updates fix known vulnerabilities that cybercriminals can exploit to gain unauthorized access or infect systems with malware.

Possible Outcome:

- Increased risk of cyberattacks
- Malware infections and system compromise
- Data breaches and service disruptions

Prevention Methods:

- Regularly install software updates and patches
- Enable automatic updates whenever possible
- Perform routine security checks on systems and devices

Part E: Security Recommendations

Security Recommendations

Security Recommendations

1. **Enable Multi-Factor Authentication (MFA)** – Adds an extra layer of security by requiring a second verification method, making it much harder for attackers to access accounts even if passwords are compromised.
2. **Use Strong Password Policies** – Encourages users to create complex and unique passwords, reducing the risk of unauthorized access through password guessing or brute-force attacks.
3. **Perform Regular System Updates and Patching** – Keeps software and operating systems protected against newly discovered vulnerabilities that cybercriminals may exploit.
4. **Provide Employee Security Awareness Training** – Educates staff about phishing, social engineering, and other cyber threats, helping to reduce human errors that can lead to security incidents.
5. **Maintain Regular Data Backups** – Ensures that important data can be restored quickly in case of ransomware attacks, accidental deletion, or hardware failures.
6. **Install and Update Antivirus Software** – Detects, blocks, and removes malware, helping to protect devices and networks from malicious software infections.
7. **Implement Access Control Measures** – Limits access to sensitive information only to authorized users, reducing the chances of data leaks or misuse.
8. **Encrypt Sensitive Data** – Protects confidential information by converting it into a secure format that cannot be easily read by unauthorized individuals.
9. **Use Firewalls and Network Security Tools** – Monitors and filters incoming and outgoing network traffic, helping to prevent unauthorized access and cyberattacks.

10. **Conduct Regular Security Audits and Vulnerability Assessments** – Identifies weaknesses in systems and networks so that they can be fixed before attackers have a chance to exploit them.

Part F: Mini Security Consultant Activity

Cybersecurity Improvement Report for College

Introduction

As educational institutions increasingly rely on digital systems for managing student records, examinations, communication, and online services, cybersecurity has become a critical requirement. This report identifies the major cybersecurity risks faced by the college and provides recommendations to strengthen its security posture.

Top 5 Cybersecurity Risks

1. **Data Breach of Student Records** – Unauthorized access to sensitive student information can lead to privacy violations and reputational damage.
2. **Unauthorized Access to Examination Data** – Attackers may modify, steal, or manipulate examination records, affecting academic integrity.
3. **Website Hacking and DDoS Attacks** – Cybercriminals may disrupt online services, making the college website unavailable to students and staff.
4. **Faculty Information Leakage** – Personal and professional information of faculty members may be exposed through phishing or data leaks.
5. **Malware and Ransomware Attacks** – Malicious software can encrypt or destroy important institutional data, causing operational disruptions.

Top 5 Recommendations

1. **Enable Multi-Factor Authentication (MFA)** for all critical accounts to prevent unauthorized access.
2. **Implement Strong Password Policies** requiring complex passwords and regular password updates.
3. **Conduct Regular Security Awareness Training** to educate students, faculty, and staff about cyber threats.
4. **Perform Regular System Updates and Security Patching** to fix vulnerabilities before they can be exploited.
5. **Maintain Secure and Regular Data Backups** to ensure quick recovery in case of ransomware attacks or system failures.

Expected Benefits

- Improved protection of student, faculty, and examination data.
- Reduced risk of cyberattacks and unauthorized access.
- Increased availability and reliability of online college services.
- Faster recovery from security incidents and data loss events.
- Enhanced trust among students, faculty, and stakeholders.

Conclusion

Cybersecurity is essential for protecting the college's digital assets and maintaining smooth academic operations. By addressing the identified risks and implementing the recommended security measures, the college can significantly strengthen its cybersecurity posture and reduce the likelihood of future cyber incidents.